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EDITORIAL

Data-driven methods in Rheology

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With a consistent growth in computational power, even today's personal computers enable storage and process of large amounts of data far beyond what was possible a decade ago. With these technological advances, over the past few years, data-driven techniques have presented themselves as transformative platforms across virtually all science and engineering disciplines. Rheology and non-Newtonian fluid mechanics are no exceptions either. Nonetheless, the breadth of topics and the pace at which the literature on data-driven methods in science and engineering is expanding makes practical use of these advances in rheology challenging. Hence, in this issue, we collect a number of articles focused on either developing or employing data-driven techniques, with the specific goal of providing the Rheologica Acta readership with robust and reliable adaptations of artificial intelligence (AI) and machine learning (ML) platforms to the rheological sciences. Needless to mention that the articles in this special issue present just a few possible ways in which advances in AI/ML methodologies can be leveraged for explorations in rheology and complex fluids, however, we hope that the articles presented here serve as a foundation for future explorations/developments in this area.

Here, we present a prelude to the eight articles collected in this special issue and provide a brief introduction to how each study leverages a machine learning technique. Just as computational methods are developed and used for different applications and different experimental techniques serve different purposes, data-driven methodologies also come in different flavors and can be adapted with particular end goals in mind. While the focal point of all eight articles in this special issue is leveraging advances in AI and ML, each of the contributions presents a distinct path for doing so. Table 1 compiles the types of data used, flow characteristics, machine learning approaches employed, and the specific applications of machine learning in each research.

Machine learning is inherently a statistical technique, and its type and application vary depending on the underlying philosophy and the quantities of interest. Within this issue, the machine learning papers covered are introduced as follows:

Weeratunge et al. (2023) propose a method utilizing Bayesian optimization, a probabilistic optimization technique forming the basis of machine learning, to search for optimal parameters in a coarse-grained polymer simulation. The study employs chain length and bond angle distributions, along with bulk density of the target polymer system, as objective functions. As a result, they demonstrate how Bayesian methodology significantly accelerates the process of determining simulation parameters. This work represents one of the most promising areas in which data-driven techniques can transform current methodologies: speedup and scale-up of numerical simulations.

Another powerful application of machine learning is in model parameter recovery based on the system variables. In their work, Farrington et al. (2023) employed machine learning techniques to represent the rheological properties, such as yield stress and viscosity of human blood as a function of its composition. The study reveals that the machine learning–based rheological model not only provides more accurate predictions than conventional models but also statistically uncovers the types and magnitudes of composition factors influencing the rheological properties, which were previously unknown. This also represents a potentially rich direction for using ML methodologies in rheology: material property predictions based on its primary constituents and characteristics.

The strength of machine learning in parameter fitting remains potent even when dealing with increased complexity in physical systems. This advantage becomes particularly prominent in solving inverse problems, where properties need to be reverse engineered from available data. Howard et al. (2023) introduce machine learning as a tool to bridge

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Table 1 Summary of the eight articles with respect to the type of data, flow type, data-driven methodology used, and application

	Data type	Flow type	Machine learning (ML) technique	Purpose of ML application
Weeratunge et al	Synthetic	Quiescent	GPR	Accelerated modeling
Farrington et al	Experimental	Couette	GPR	Parameterization (CE)
Howard et al	Synthetic	Pressure driven	Galerkin	Micromechanics
Lennon et al	Synthetic + experimental	Couette, MAPS	MLP	Parameterization (MAPS)
Dabiri et al	Synthetic	Couette	RhINN (PINN)	CE development
Jin et al	Synthetic	Couette	RNN	CE development
Young et al	Synthetic	Complex	PCA, neural ODE	Fluid/flow modeling
Kang et al	Experimental	Pressure driven, chaotic	ESN	Material classification

GPR Gaussian process regression, *CE* constitutive equation, *RNN* recurrent neural network, *ESN* echo state network, *MLP* multilayer perceptron, *MAPS* medium amplitude parallel superposition, *PINN* physics-informed neural network, *ODE* ordinary differential equation

between microscopic variables and transport equations in colloidal suspension simulations using the force coupling method. In the macroscopic world, transport equations are defined as relationships between statistically averaged bulk properties such as concentration, stress, and velocity. On the other hand, particle simulations provide individual particle behavior separately, leading to significant heterogeneity in locally calculated properties. In this research, machine learning demonstrates its capability to extract macroscopic variables and transport coefficients by learning the micromechanics of the problem at hand.

Lennon et al. (2023) and Dabiri et al. (2023) also employ machine learning in solving inverse problems. In Lennon et al.'s study, they focus on the medium amplitude parallel superposition (MAPS) rheology of the soft glassy rheology (SGR) model. Under MAPS flow conditions, the third-order complex modulus, G_3^*, can be defined. However, G_3^* of the SGR model is a complex function of the model parameters, making it rarely possible to extract SGR model parameters from MAPS experimental data using traditional methods. In contrast, this study introduces machine learning techniques to efficiently solve the inverse problem. In the case of Dabiri et al., they work with the fractional viscoelastic model (FVEM) and construct rheology-informed neural networks (RhINNs) based on the physics-informed neural network (PINN) framework. Fractional constitutive models present an efficient way of representing a constitutive relation with the most compact form possible; nonetheless, deriving and/or solving fractional constitutive models are far from trivial. Thus, methods that facilitate construction or solution of fractional differential equations to describe a set of physical properties can change the way these extremely efficient classes of models are used. The fractional RhINNs model developed in Howard et al. (2023) using relaxation modulus data demonstrates that this approach allows for straightforward extraction of FVEM parameters from data, even in cases where analytical fitting methods struggle. These studies can be considered significant contributions to the field of rheology, addressing important topics related to constitutive equations and rheological properties.

Jin et al. (2023) and Young et al. (2023) focus on developing constitutive equations for complex fluids based on data. In Jin et al.'s study, the authors propose a constitutive equation based on a recurrent neural network (RNN). They design a recurrent unit to have a fading memory with spectrum like a rheological fluid. Through this approach, they demonstrate that a single neural network structure can learn the behaviors of different complex fluids without the need for pre-defined governing equations unlike most PINNs. Young et al. design a system to observe the velocity, stress, spatial distribution of microstructures, and their temporal changes in complex flows, including shear and elongation. From the trajectory of the fluid obtained from this system, they implement the representation of structural factors and stress evolution under the flows using the Neural ODE technique, which involves solving and training differential equations. Additionally, they use principal component analysis and autoencoder techniques to compress high-dimensional image data, specifically scattering intensity, into a reduced set of structural information. As a result, they not only develop the closed-form neural network-based constitutive equations from the observed data but also demonstrate the ability to track the evolution of the latent state which is directly related to the microstructure. Both studies contribute to the development of constitutive equations for complex fluids by utilizing data-driven approaches, showcasing the potential of machine learning in understanding and predicting the behaviors of complex fluid systems.

The previously mentioned studies all fall under the category of regression problems, where machine learning is used to predict clearly defined physical quantities. However, another type of optimization problem, namely classification, can also be effectively used in the field of rheology. Kang et al. (2023) observe real-time pressure and flow signals

during the transport of battery slurries with varying dispersion and composition. The signals exhibit random or chaotic oscillation patterns over time, which makes it difficult for traditional signal processing methods to distinguish and classify the signals from different slurries. In this research, they demonstrate that by utilizing an echo state network, it is possible to discern the oscillation patterns of different signals. This not only shows that the physical characteristics of complex fluids are reflected in temporal fluctuation during the flow but also highlights the potential of machine learning as a useful tool for real-time diagnosis of fluid states.

Declarations

Competing interests The authors declare no competing interests.

References

Dabiri D, Saadat M, Mangal D et al (2023) Fractional rheology-informed neural networks for data-driven identification of viscoelastic constitutive models. Rheol Acta. https://doi.org/10.1007/s00397-023-01408-w

Farrington S, Jariwala S, Armstrong M et al (2023) Physiology-based parameterization of human blood steady shear rheology via machine learning: a hemostatistics contribution. Rheol Acta. https://doi.org/10.1007/s00397-023-01402-2

Howard et al (2023) Machine learning methods for particle stress development in suspension Poiseuille flows. Rheol Acta. https://doi.org/10.1007/s00397-023-01402-2

Jin H, Yoon S, Park FC et al (2023) Data-driven constitutive model of complex fluids using recurrent neural networks. Rheol Acta. https://doi.org/10.1007/s00397-023-01405-z

Kang S, Jin H, Ahn CH et al (2023) Classification of battery slurry by flow signal processing via echo state network model. Rheol Acta. https://doi.org/10.1007/s00397-023-01404-0

Lennon KR, Rathinaraj JDJ, Gonzalez Cadena MA et al (2023) Anticipating gelation and vitrification with medium amplitude parallel superposition (MAPS) rheology and artificial neural networks. Rheol Acta (2023). https://doi.org/10.1007/s00397-023-01407-x

Weeratunge H, Robe D, Menzel A et al (2023) Bayesian coarsening: rapid tuning of polymer model parameters. Rheol Acta. https://doi.org/10.1007/s00397-023-01397-w

Young CD, Corona PT, Datta A et al (2023) Scattering-Informed Microstructure Prediction during Lagrangian Evolution (SIMPLE)—a data-driven framework for modeling complex fluids in flow. Rheol Acta. https://doi.org/10.1007/s00397-023-01412-0

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